

Chapter 10

Developing the Exercise Program

The exercises described in this chapter constitute a program founded on the principles discussed in previous chapters. This program is sufficient for many patients and for those interested in optimal back health. Those with performance ambitions are directed to my textbook *Ultimate Back Fitness and Performance* (McGill, 2014); also, chapter 11 offers some insight into this different world. Although I claim no expertise beyond the low back, I will broach other issues that have implications for the back. For example, improper technique when performing hip and knee mobilization and stretching can compromise the back; for this reason, I address these issues here. Upon completion of this chapter, you will be able to design exercise that challenges muscle and establishes stabilizing motor patterns, but spares the spine. The focus here is clearly on the person with back pain and on low back health for everyday activity.

Before presenting the exercises, I review some of the basic ideas discussed in this book thus far, such as stretching without stressing the back and abdominal hollowing and bracing. I also introduce the technique of flossing the nerve roots for those with sciatica.

Philosophy of Low Back Exercise Design

Many traditional notions that exercise professionals consider to be principles for exercise design, particularly when dealing with the low back, are not as well supported with data as generally thought. As previously noted, many clinicians still prescribe sit-ups—usually with the proviso that the knees remain bent. The resultant spinal loading of well over 3,000 N of compression to a fully flexed lumbar spine clearly shows the folly of such a recommendation. Other experts still recommend that people adopt a posterior pelvic tilt when performing many types of low back exercise. This actually increases the risk of injury by flexing the lumbar joints and loading passive tissues. The recommendation of flattening the lumbar region to the floor when performing abdominal exercise is another version of this ill-founded philosophy. Many continue to believe that having stronger back and abdominal muscles is protective and reduces bad back episodes, even though Luoto and colleagues (1995), among others, showed that muscle endurance, not strength, is more protective. This should not be misinterpreted to mean that having stronger muscles is not a good objective; rather, it reflects on many of the strength training approaches currently used that create back patients. Myths remain regarding the need for greater lumbar mobility, which the evidence suggests leads to more back troubles—not fewer (e.g., Biering-Sorensen, 1984)! Finally, we are disturbed by the fact that replicating the motions and spine loads that occur while using many low back extensor machines, intended as training and therapy, produced disc herniations when applied to spines in our laboratory!

Clinical Wisdom

Clearly, some current clinical wisdom needs to be reexamined in the light of the scientific evidence, much of which has been presented in this book. Removing the cause of pain is essential. Spine and torso stiffness enhances performance and reduces the risk of injury—this is unequivocal. The evidence indicates that the safest and most mechanically justifiable approach to enhancing lumbar stability through exercise is to emphasize endurance over strength. Clinicians using such an approach should encourage patients to maintain a neutral spine posture when under load and use abdominal cocontraction and bracing in a functional way. Studies of military personnel show this to be effective (e.g., Suni et al., 2013), as do large studies of people with back pain versus matched controls without pain. People with back pain move in ways that stress the back, ensuring pain (Luomajoki et al., 2008)! The bodybuilding approach often used in rehabilitation is not synonymous with health objectives and results in additional risk from joint loading through a range of motion. Further, the simple isolationist motor patterns do not promote pain-reducing stability.

Thoughts on Mobility

This book has introduced issues regarding spine mobility, whether stretching is appropriate or effective, and the consequences of pain on mobility. The next few sections guide the application of scientific findings on mobility and moving with competence.

Sparing the Back While Stretching the Hips and Knees

Given that tonic contractions in the psoas, which produce a shortening and hip extension restriction, are common in those with back troubles (McGill et al., 2003), people with this condition would probably benefit from hip stretching and other range of motion exercises. Sometimes, however, poor technique during the stretching of hip and knee tissues leads to unnecessary loading of the lumbar spine. This can easily be avoided by changing technique. A general guideline for sparing the back is to maintain an upright torso posture while performing hip and knee work. Preservation of a neutral spine ensures minimal loading from passive tissues. An upright torso minimizes the reaction torques, the associated muscle contraction, and spine load.

Abdominal bracing to an appropriate level is sometimes required for pain control. For example, lunges are a good exercise for challenging strength, endurance, balance, and mobility in the lower extremities. Lower-extremity capability is needed to facilitate spine-sparing postures when lifting and when performing a host of other tasks.

The technique for sparing the back is to maintain an upright torso while performing the lunge (see [figure 10.1](#)). Some clinicians recommend keeping the back leg straight, which causes patients to flex the torso forward. This is poor form for sparing the back (see [figure 10.2](#)). A general principle is to keep the torso upright with a neutral lumbar spine while stretching joints other than the back. [Figure 10.3](#), a and b, for example, shows spine-sparing postures for stretching the quads and the hip adductors, respectively. Finally, the lunge with a neutral spine stretches the iliopsoas more than the psoas (the iliopsoas is a uniarticular muscle that crosses only the hip). Fully stretching the psoas requires a lateral bend of the torso away from the extended hip. The

anatomy train is then engaged by raising the same-side arm and pushing the hand to the ceiling, but this is reserved for those with more robust backs (see [figure 10.4](#), a-d).

Figure 10.1 The spine is spared performing the beginner's lunge by maintaining an upright torso and neutral spine curvature.



Figure 10.2 Some clinicians recommend keeping the back leg straight while performing lunges, which causes patients to flex the torso forward. This is poor form for sparing the back.



Figure 10.3 During stretching at joints other than the back, the spine is spared by maintaining an upright torso and neutral spine curve. (a) When stretching the quads, holding a chair for balance is a good way to ensure a straight back. (b , c) This person has sufficient hip flexion mobility to ensure an upright torso and a neutral spine. Those who are unable to maintain this spine posture should forgo this exercise until they can achieve the required hip flexion mobility. (d) This shows poor technique with spine motion.

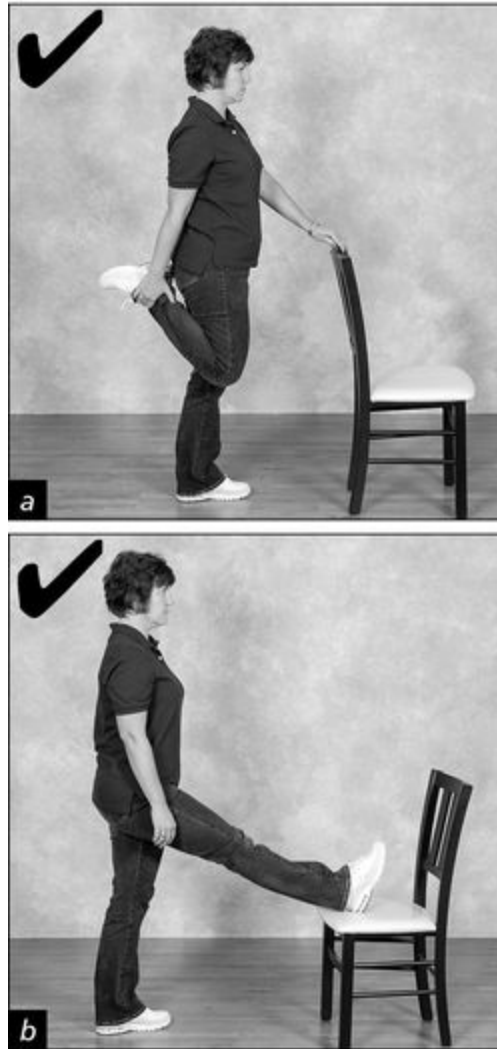




Figure 10.4 Unlike the uniarticular iliacus, the multiarticular psoas crosses the hip and entire lumbar spine, and needs specific techniques to stretch it. (a) The patient will feel the psoas tendon just medial to the rectus femoris, below the inguinal ligament. (b) Then they descend into a lunge, (c) noting the absence of tension in the psoas tendon. The ipsilateral arm is raised and pushed straight up. (d) The psoas should now be noticeable. Leaning the torso away from the arm pushing up will increase the psoas tension.





Flossing the Nerve Roots for Those With Accompanying Sciatica and Neural Tensions

Before presenting the flossing technique for sciatica, some background is required regarding the biomechanics and biochemistry of the spinal cord and nerves.

Excessive tension in the nerve is the cause of sciatica. Normally, the cord nerve roots travel through the foramens of the neural arch of each vertebra, and their sliding excursions are substantial with spine flexion, hip flexion, and knee extension. According to Louis (1981), the nerve can travel well over a centimeter (over 1/2 in.) during some of these movements. Thus, if a

nerve root is impinged and cannot slide, any of the postures that would normally pull that nerve root through the foramen will increase nerve tension. Instead of moving, the nerve is stretched. Tension on such a nerve can be increased from the cranial end with simultaneous cervical flexion because the entire spinal cord moves slightly with cervical flexion and thus pulls at the nerve roots all along their length. This is the basis for the tests for nerve tension presented in chapter 9.

Recently, scientists have suggested that nerves have the ability to create their own pathways as long as they can move. They seem to have some chemically based ability to dissolve—over time—tissues impinging them, although the exact molecular mechanism remains elusive. The idea of flossing is to pull the cord and nerves from one end only, while releasing at the other, and then to switch the pull-and-release direction. In this way the nerve roots are flossed through the lateral foramina and in fact along their entire length. Thus, by working the nerves back and forth in whatever limited range they can manage in spite of impingement, we facilitate the dissolving of the impingement and the gradual release of the nerve to once again move freely (this approach was proposed by Butler, 1999). This flossing action is accomplished with coordinated hip, knee, and cervical motion.

A note of caution is needed here: Although this can be wonderful in helping chronic sciatica resolution, it can also cause an acute onset. Be very conservative in the first session. If the nerve is adhered so that it can't slide, the sciatic symptoms will be exacerbated. However, if the patient reports no change or even relief the next day, then increase the flossing. Monitor all patients and remove the procedure from the programs of any whose symptoms worsen.

Flossing

The patient, seated and with the legs able to swing freely (a), flexes the cervical spine while at the same time flexes the knee and plantar flexes the ankle (b). This creates a pull on the spinal cord from the cranial end and a release from the caudal end. This should not produce sciatic symptoms (see the discussion of the supine passive leg raise test in chapter 9). Then, the patient extends the cervical spine with simultaneous knee extension on the side with sciatica (c, d). This pulls the nerve from the caudal end with a

corresponding release at the cranial end. The cycle is completed as the knee is flexed with coordinated motion of the cervical spine flexing. This causes the nerves to floss through the vertebral tissues. If the patient experiences minor sciatic symptoms when the cervical spine is flexed or the knee is extended, then the patient should reduce the range of motion at these joints until no pain is provoked. Some patients who have suffered sciatica pain for years report reductions in their sciatic symptoms within a few days to a couple of weeks; others report increasing symptoms.







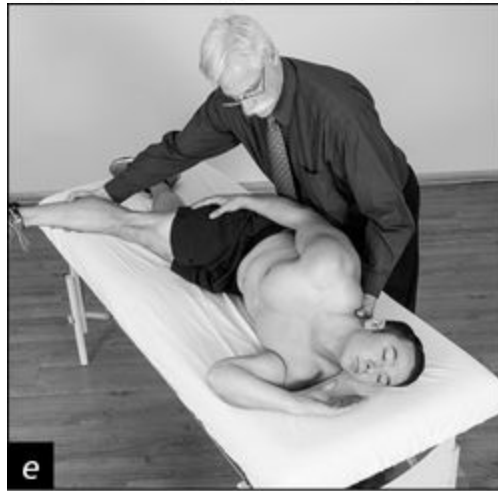
Technique Tips

- A tip to enhance efficacy is to first adopt a posture that relieves impingement. For example, the patient with a disc bulge who can reduce the bulge by adopting a prone extension posture (such as shown in the McKenzie posture test in chapter 9) should do so for a few minutes prior to flossing. The patient should then adopt the posture with the longest reflex response.
- The motion should be slow (with a flexion–extension cycle lasting about 5 seconds), but most important, coordinated, to create floss rather than length change in the neural tract.
- The objective is to create motion rather than a static stretch. The patient should not push into the end range, and never travel into a painful range.
- The patient should begin by performing 10 repetitions with each leg. Then, if symptoms are not exacerbated, he can perform these repetitions several times per day. The patient should not perform flossing within 2 hours after rising from bed.

Advanced Flossing

Recent work of Professor Olavc Airaksinen and colleagues (personal communication, June 2013) showed that double-leg flossing increased the central neural tract excursion distance and may be considered as an exercise to add to the routine (a, b). The progression continues in side-lying in which the hip joint becomes involved in creating large nerve excursions. The neck is flexed while the hip and knee are extended, and the ankle is plantar flexed (c). The cycle is completed with the opposite movement (d, e).





Identifying Safe and Effective Exercises

After a brief review of the muscles that stabilize the spine, we will have a look at exercises for training those muscles. Although many muscles have been regarded as primary spine stabilizers, the confirmation of their roles requires two levels of analysis. The first is engineering—a stability analysis must be conducted on anatomically robust spine models to document the ability of each component to stiffen and stabilize. Second, electromyographic recordings of all muscles (even deep muscles requiring intramuscular electrodes) are necessary to confirm the extent to which the motor control system involves each muscle to ensure sufficient stability.

Identifying safe and effective exercises for low back stabilization is a several-step process: incorporating and patterning the muscles involved in spine stability; eliminating unsafe exercises; and selecting or devising exercises that appropriately challenge the selected muscles at the various stages of rehabilitation and conditioning.

Incorporating and Patterning the Muscles

Our intramuscular and surface electromyography (EMG) and modeling studies to quantify spine stability demonstrated that virtually all torso muscles play a role in stabilization. (Some data are presented in chapter 4.) Although the back extensors and the abdominal wall all appear to play significant roles, the quadratus lumborum also appears to play an important role in many tasks. The fibers of quadratus lumborum cross-link the vertebrae; they have a large lateral moment arm via the transverse process attachments; and they traverse between the rib cage and iliac crests. In this way, the quadratus is an effective stiffener and appears to be effective in all loading modes, as a result of its architectural design. Typically, the first mode of lumbar buckling is lateral; the quadratus appears to play a significant role in local lateral buttressing. The three layers of the abdominal wall are also important for stability together with the muscles that attach directly to vertebrae—the multisegmented longissimus and iliocostalis and the unisegmental multifidii. Cholewicki and McGill (1996) also presented an argument for the role of the small intertransversarii in producing small but

critical stabilizing forces. As well, latissimus dorsi plays a major stabilizing role in many loading tasks (Kavicic, Grenier, and McGill, 2004). On the other hand, psoas activation appears to have little relationship with low back demands; the motor control system activates it when hip flexor moment is required (see Andersson et al., 1996; Juker et al., 1998), limiting its role in stabilizing the spine when hip flexion is not involved. But when corresponding hip flexion is demanded, the psoas becomes an important spine stabilizer via shear stiffness, compression, and locking the pelvis to the lumbar spine.

Eliminating Unsafe Exercises

Quantitative data have confirmed that no single abdominal exercise challenges all of the abdominal musculature while sparing the back (Axler and McGill, 1997). For this reason, more than one exercise is required. Unfortunately, many inappropriate exercises have been prescribed for people with low back difficulties, including the following:

- Sit-ups (both straight-leg and bent-knee) are characterized by higher psoas activation, with consequent high low back compressive loads that exceed the occupational guidelines of the National Institute for Occupational Safety and Health (NIOSH) in the United States.
- Leg raises cause even higher psoas activation and spine compression (actual values are presented in chapter 4).
- Most traditional extensor exercises are characterized by very high spine loads, which result from externally applied compressive and shear forces (from either free weights or resistance machines). A commonly prescribed spine extensor muscle challenge, the superman, involves lying prone while extending the arms and legs (see [figure 10.5](#)). This results in over 6,000 N of compression to a hyperextended spine, transfers load to the facets, and crushes the interspinous ligament. Needless to say, this exercise is contraindicated for anyone at risk of low back injury—or reinjury! Although some may believe that putting the hands on either side of the head rather than extending them may make this exercise safe, this is not true. This exercise should not be done in any form. Further, recall that the mechanism for disc herniation is reproduced by back machines that take the lumbar spine from full

flexion and through the range of motion under load from muscle contraction.

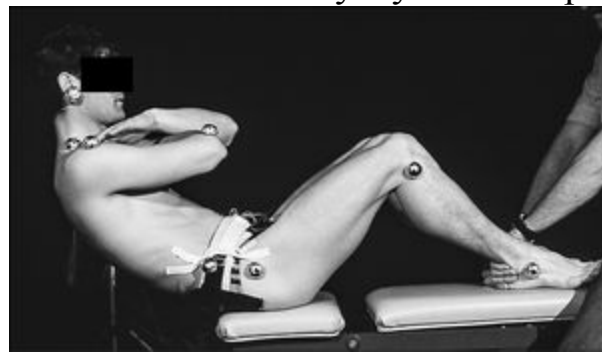
Figure 10.5 A commonly prescribed spine extensor muscle challenge (superman) involves lying prone while extending the arms and legs. This results in over 6,000 N of compression to a hyperextended spine. It is a poorly designed exercise for people with back pain.



Photos from Stuart McGill

In our investigations we made several relevant observations regarding psoas activation during abdominal exercises. The challenge to the psoas is lowest during curl-ups, followed by higher levels during the horizontal side bridge. Bent-knee sit-ups were characterized by larger psoas activation than straight-leg sit-ups, and the highest psoas activity was observed during leg raises and hand-on-knee flexor isometric exertions. The often-recommended press-heels sit-up, which has been hypothesized to activate the hamstrings and neurally inhibit the psoas, was actually confirmed to increase psoas activation! (See [figure 10.6](#).) (Original data can be found in Juker et al., 1998; clinicians and coaches who intentionally wish to train the psoas will find these data informative.) Once again, the horizontal side bridge appears to have merit because it challenges the lateral obliques and transverse abdominis without high lumbar compressive loading and ensures a high stability index (a loss of contraction of the involved muscles would cause the patient to fall out of the bridge).

Figure 10.6 The press-heels sit-up was proposed by several clinical groups, on theoretical grounds, to inhibit the psoas by activating the hamstrings. In fact, EMG assessment (Juker et al., 1998) proved this to be mythical. This original photo from the intramuscular experiments shows the clinician's hands behind the heels of the subject as the hamstrings are being activated during the sit-up. Activating the hamstrings creates a hip extensor moment, and sit-ups require hip flexion. During this type of sit-up, the psoas is activated to even higher levels to overcome the extensor moment from the hamstrings and produce a net flexor moment. This type of sit-up produced the highest level of psoas activation of any style of sit-up we quantified!



Photos from Stuart McGill

Upper and Lower Rectus Abdominis?

Myoelectric evidence, normalized and calibrated, suggests that there is no functional distinction between an upper and lower rectus abdominis in most people; in contrast, the obliques are regionally activated with upper and lower neuromuscular compartments as well as medial and lateral components. There are, however, some highly trained people who can create small differences in activation. However, these differences are only at very low levels of activation and occur during what would be considered nonfunctional tasks, such as belly dancing (Moreside, Vera-Garcia, and McGill, 2008).

Selecting Safe and Effective Exercises

A wise choice for stabilization exercises in the early stages of training or

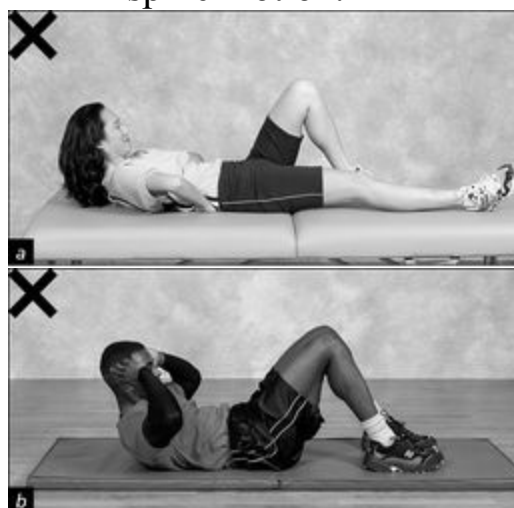
rehabilitation, and for simple low back health objectives, is the big three that we have quantified to sufficiently challenge muscle, spare the spine of high load, and ensure sufficient stability:

- Curl-ups for rectus abdominis
- Several variations of the side bridge for the obliques, transverse abdominis, and quadratus
- Leg and arm extensions leading to progressions of the bird dog for the many back extensors

The variation of each of these exercises must be chosen with the patient's or athlete's status and goals in mind.

- Curl-up: training rectus abdominis. Calibrated intramuscular and surface EMG evidence suggests that the various types of curl-ups challenge mainly the rectus abdominis because psoas and abdominal wall (internal and external oblique and transverse abdominis) activity is relatively low (see the tables in chapter 4 for relative activation levels in a variety of exercise tasks). Curl-ups performed with poor technique, however, can be counterproductive, either failing to activate the rectus abdominis sufficiently or overstressing the spine (see [figure 10.7](#), a and b). Curl-ups with a twisting motion are expensive in terms of lumbar compression because of the additional oblique challenge. Higher oblique activation with lower spine load is accomplished with the side bridge, which is therefore preferred over twisting curl-ups for training the obliques. The side bridge is presented next. The highest-level curl-up is presented in chapter 11.

Figure 10.7 (a) Poor form during the curl-up is to flex the cervical spine, loading the neck and not the rectus. (b) Another common type of poor form is to elevate the head and shoulders a large distance off the floor. This patient is elevating far too much, which is closer to replicating the much higher stresses of the sit-up. The intention is to activate rectus and not to produce lumbar spine motion.



- Side bridge : training the quadratus lumborum , lateral obliques , and transverse abdominis. Given the architectural and EMG evidence for the quadratus lumborum, transverse abdominis, and abdominal obliques as spine stabilizers, the optimal technique to maximize activation but minimize spine load appears to be the side bridge. Abdominal bracing is emphasized in all forms of this exercise. Maintaining the bridge ensures constant muscle activation, and the brace introduces new combinations of muscle recruitment to ensure stability. It is almost impossible for the spine to become unstable during the performance of a side bridge with a neutral spine.
- Bird dog: training the back extensors. In our search for methods to activate the extensors (including longissimus, iliocostalis, and multifidii) with minimal spine loading, we have found that the single-leg extension creates tolerable spine loading ($<2,500$ N) for many and activates one side of the lumbar extensors to approximately 18% of maximal voluntary contraction (MVC). Simultaneous leg extension with contralateral arm raise (the bird dog) increases the unilateral extensor muscle challenge (approximately 27% of MVC in one side of the lumbar

extensors and 45% of MVC in the other side of the thoracic extensors) but also increases lumbar compression to well over 3,000 N (Callaghan, Gunning, and McGill, 1998). This exercise can be enhanced with abdominal bracing and deliberate mental imaging of activation of each level of the local extensors. Once again, the technique for challenging the extensors should be guided by the patient's status and goals. This is an individual clinical decision.

Beginner's Program for Pain Control and Stabilization

Now that you are familiar with some specific recommended low back exercises, how do you put these together to create the best possible program for your patient? Assuming that there were no particular indicators obtained during provocative testing that would contraindicate beginning with a typical progression, we recommend considering the following sequence:

1. Throughout the day incorporate the movement tools to eliminate tissue irritation. Irritation sensitizes the tissues to trigger pain at lower levels of stimulus. (Recall chapter 8.)
2. Begin the training session with the flexion–extension cycles, also called the cat–camel exercise (see [figure 10.8](#), a and b), to reduce spine viscosity. Note that the cat–camel is intended as a motion exercise, not a stretch, so the emphasis is on motion rather than pushing at the end ranges of flexion and extension. We have found that five or six cycles are often sufficient to reduce most viscous stresses—additional cycles rarely reduce viscous friction further. Those with sciatica may find increased symptoms during the flexion phase. Use pain to guide the choice of a suitable pain-free range of motion.
3. Perform slow partial squats in which the pelvis is directed downward and posterior along a 45° line. The spine has no motion (see the description of the potty squat in chapter 8).
4. Sciatic pain sufferers may try the nerve-flossing technique following a posture that relieves neural tension.
5. These motions are followed by anterior abdominal exercises—namely, appropriate curl-ups.
6. Lateral musculature exercises follow the preceding—namely, the side bridge for quadratus lumborum and the muscles of the abdominal wall, for optimal stability.
7. The extensor program begins in the quadruped position with single thigh extensions progressing to the full bird dog.
8. The person begins a walking program based on tolerable intervals. If she can walk only 10 meters (or yards) without pain, then she should walk 7 meters every hour. As she gains tolerance, increase the distance and the time between intervals. Always design the distance to be less than the

distance at which pain would occur.

Figure 10.8 Cat–camel exercise. Note that this is a motion exercise and not a stretch; the person should not push at the end range of motion. Viscosity is measurably reduced after just a few cycles.



Photos from Stuart McGill

In general, we recommend that the isometric holds performed in the curl-up, bridge, and bird dog be no longer than 7 or 8 seconds, given recent evidence from near-infrared spectroscopy indicating rapid loss of available oxygen in torso muscles contracting at these levels. Short relaxation of the muscle restores oxygen (McGill, Hughson, and Parks, 2000). The endurance objectives are achieved by building up repetitions of the exertions rather than by increasing the duration of each hold.

Motivated by evidence of the superiority of extensor endurance over strength as a benchmark for good back health, we documented normal ratios of endurance times for the torso flexors relative to the extensors and lateral musculature (see chapter 9). Use these values to identify endurance deficits—both absolute values and for one muscle group relative to another—and to establish reasonable endurance goals for your patients.

A Crucial Note to Clinicians

Clinical Relevance

Insistence on perfect technique will allow the patient to reach much higher levels of challenge without pain (Ikeda and McGill, 2012; McGill and Karpowicz, 2009). Do not allow spine deviation such as lumbar flattening to the floor or loss of neutral during holding of any stabilizing posture. Constant postural correction is necessary in some patients. Finding the optimal dosage of load is also critical. Keep correcting to eliminate any pain.

Exercises That May Be Used in a Stabilization Program

The basic big three exercises (an appropriate form of the modified curl-up, side plank, and bird dog) have been found to be the most effective in working with patients with low back troubles.

Beginner Through Advanced Curl-Ups

This series of curl-ups, along with neck-strengthening isometric exercises for the neck (as necessary), provides a good foundation for a strong rectus abdominis.



Curl-Up, Beginner

The curl-up technique is critical to spare the spine. The basic starting

posture is supine with the hands supporting the lumbar region. The person should not flatten the back to the floor, which takes the spine out of elastic equilibrium and raises the stresses in the passive tissues. Although the position of elastic equilibrium is desired in the lumbar region, the hands can be adjusted to minimize pain if needed. One leg is bent with the knee flexed to 90° and the other leg remains relaxed on the floor. This adds further torque to the pelvis to prevent the lumbar spine from flattening to the floor. The focus of the rotation is in the thoracic spine; many tend to flex the cervical spine, which is poor technique. Rather, picture the head and neck as a rigid block on the thoracic spine. No cervical motion should occur—either chin poking or chin tucking. The intention is to activate rectus and the obliques and not to produce spine motion. People who report neck discomfort may try the isometric exercises for the neck that follow. In addition, particularly for patients experiencing neck discomfort, the tongue should be placed on the roof of the mouth behind the front teeth and pushed upward, which helps to promote stabilizing neck muscle patterns. Patients should leave the elbows on the floor while elevating the head and shoulders a short distance off the floor. The rotation is focused in the midthoracic region. The head and neck unit is locked onto the rib cage.



Photos from Stuart McGill

Shown in the photos are (a) a very mild form of the curl-up to just take the weight off the head and shoulders with almost no motion. (b) More challenge is obtained with raising the head and shoulders but focusing the motion on the thoracic spine with no cervical or lumbar flexion.

Isometric Exercises for the Neck

Those who experience neck symptoms with curl-ups may find relief by

building the neck by developing better control (software) than muscular strength (hardware). Begin these isometric exercises where the head and neck unit does not move, and the tongue is on the roof of the mouth behind the front teeth and pushed upward to activate the deep flexor muscles (a). The fists are under the chin where they push up (b). The neck maintains isometric contraction to ensure that no motion occurs. The patient holds for several seconds, then relaxes, building up endurance and grooving stabilizing motor patterns by increasing the repetitions of the hold-relax cycles. (Believe it or not, some patients try this while chewing gum, which makes the establishing of stabilizing patterns impossible.) More traditional isometric exercises (c-e) may or may not help.





Curl-Up, Intermediate

The intermediate progression of the curl-up involves raising the elbows a couple of centimeters (a little less than an inch) so that the arms do not pry the shoulders up, thus shifting more load to the rectus. The person should not raise the head and neck any higher than in the beginner curl-up.



Curl-Up, Advanced

An advanced curl-up requires prebracing of the abdominal wall. The resistance is provided by the brace, and no additional motion takes place. This can be facilitated and learned with fascial raking by a clinician. Deep breathing is also added to train the diaphragm to function independently of the stabilizing abdominal musculature. The head and neck must move as a unit, maintaining their rigid-block position on the thoracic spine.



The side bridge is an excellent exercise to challenge the important stabilizers of quadratus lumborum, latissimus dorsi, and the abdominal obliques, while sparing the spine from high loads. Although there are several forms that comprise a logical progression, each form is characterized by a neutral spine with the rib cage locked to the pelvis.

Side Bridge, Remedial

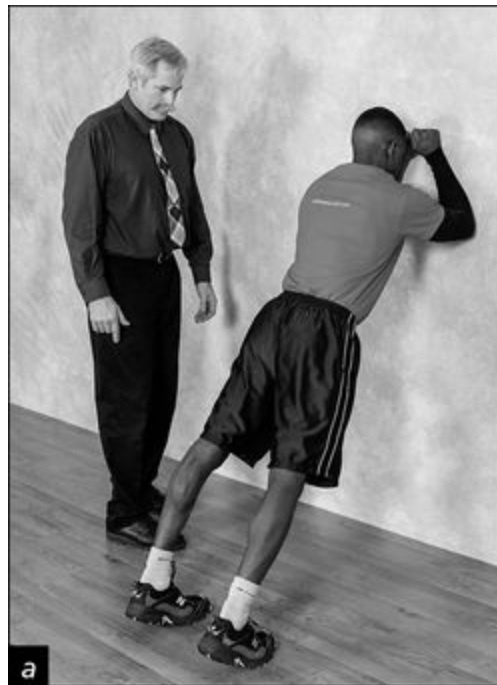
Many patients deserve special consideration—for example, the chronic patient who is quite deconditioned. These patients are sometimes unable to

perform the side bridge even from the knees. Start these people with a side bridge while standing against a wall. Instruct them to move smoothly from the beginning position (a) through the intermediate position (b) to the final position (c), in one flowing motion. The patient should pivot over the toes, as if performing a fluid dance move. Clinical cues may include feedback such as keeping the heels of the hand on the pelvis and the fingertips on the rig cage to eliminate lumbar motion.

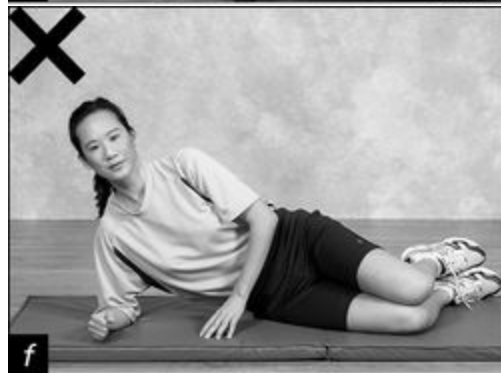
Once patients graduate to the floor, they can spare the spine by beginning (d) with the knees and hips quite flexed and move into the side bridge posture with an accordion-like unfolding of the legs (or a hip-hinge squat; d, e), all the time keeping a braced neutral spine. Never allow the patient to (f) slump into a laterally bent, deviated spine posture.

Another type of patient is the football player whose shoulders are so painful he cannot tolerate the shoulder load (some elderly women also fall into this patient group). These patients can perform a modified side bridge by lying on the floor and attempting to raise the legs laterally (g, h) or simply attempting to take the weight off the legs.

Another option for patients with shoulders that cannot tolerate load is to stand on a 45° bench with the feet anchored, which spares the shoulders (j-l).











Photos from Stuart McGill (photos d, e, g-l)

Side Bridge, Beginner Through Advanced

Once the client is comfortable performing the final remedial side bridge, she may move on to the beginner through advanced side bridge progression.

Side Bridge, Beginner

Beginners bridge from the knees. In the beginning position, the exerciser is on the side, supported by the elbow and hip. The knees are bent to 90°. Placing the free hand (the hand and fingers are spread and cap over the deltoid) on the opposite shoulder and pulling down robustly on it will help stabilize the shoulder. The torso is straightened until the body is supported on the elbow and the knee, with some input from the lower leg. The beginner side bridge can be slightly advanced by placing the free arm along the side of the torso—effectively placing more load on the bridge.



Photos from Stuart McGill

Side Bridge, Intermediate

The beginning position for the intermediate side bridge is like that for the beginner, except that the legs are straight. The torso is straightened until the body is supported on the elbow and feet (see photo). When supported in this way, the lumbar compression is a modest 2,500 N, but the quadratus closer to the floor appears to be active up to 50% of MVC (this is a preferred exercise for the obliques because they experience similar levels of activation).



Side Bridge, Intermediate Variation

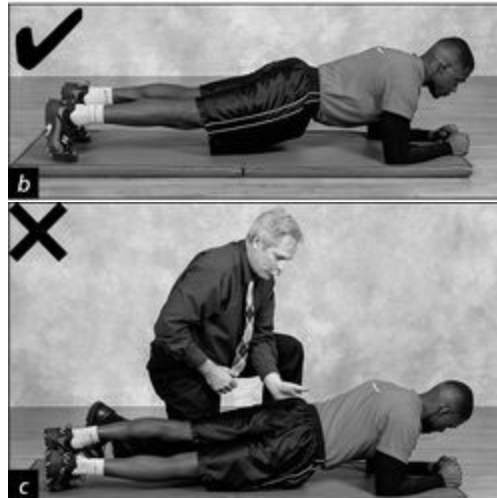
Placing the upper leg and foot in front of the lower leg and foot enables a longitudinal rolling of the torso to challenge both anterior (a) and posterior (b) portions of the abdominal wall.



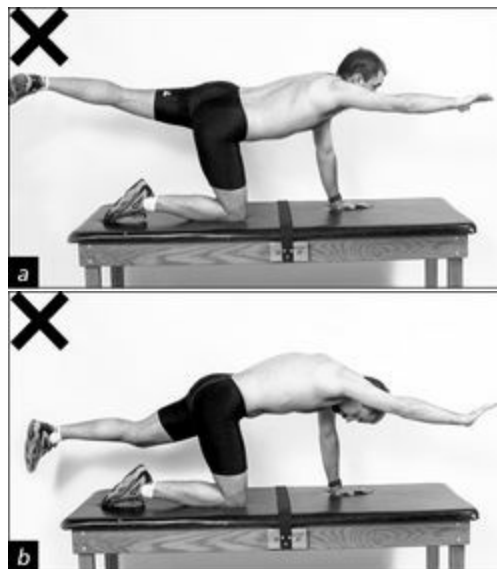
Side Bridge, Advanced

The advanced technique to enhance the motor challenge of the side bridge requires transferring from one elbow to the other while abdominally bracing (a-c) rather than repeatedly hiking the hips off the floor into the bridge position. Ensure that the client's rib cage is braced to the pelvis and that this rigidity is maintained through the full roll from one side to the other (a, b). You may cue the patient by placing the heel of your hand on the iliac crest with your fingers touching the rib cage. This ensures that the pelvis does not lead the rib cage during rolling and makes the exercise more challenging and safer for the spine. Poor form due to spine twist is shown in c. Still higher levels of activation would be reached with the feet on a labile surface (Vera-Garcia, Grenier, and McGill, 2000). This technique is reserved for the athletic back.





Remember to emphasize abdominal bracing and a neutral spine throughout all versions of the final of the big three exercises—the bird dog. Poor form includes hip hiking or any other configuration that causes deviation (twist, flexion, or lateral bending) to the spine (see a and b).



Common mistakes made in the bird dog are (a) hiking the hip, which twists the spine, and (b) not achieving a neutral spine. A flexed posture is shown here.

Photos from Stuart McGill

Remedial Through Advanced Bird Dog

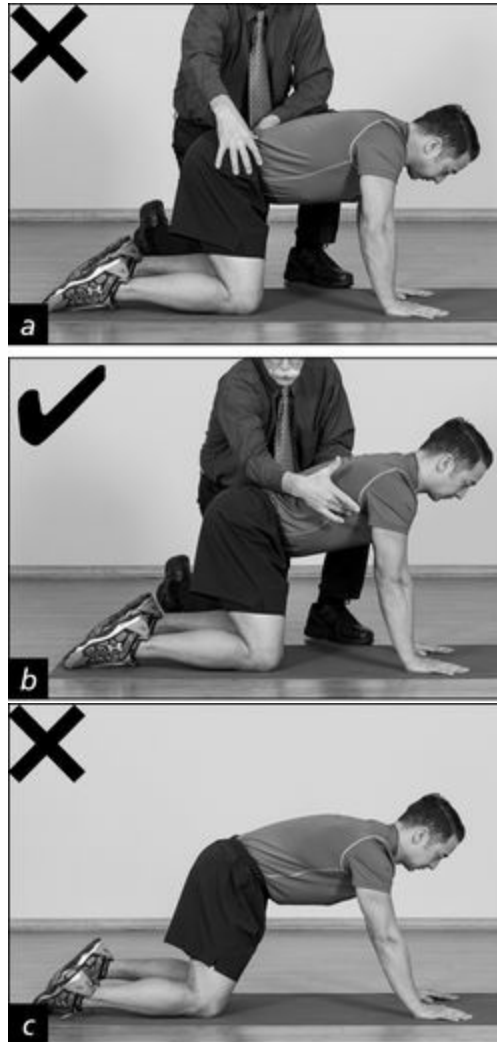
Recall that the basic bird dog trains the extensor muscles, even though the abdominals are consciously activated to control and maintain the neutral spine posture. As always, progression based on each patient's unique needs and responses to exercise, along with insistence on correct form, is indispensable to ensuring that the patient experiences strength and endurance gains without injury.



Bird Dog, Remedial

The starting position is on the hands and knees with the hands under the shoulders and the knees directly under the hips (a). For the patient with a very deconditioned back, this exercise involves simply lifting a hand or knee about an inch (2.5 cm) off the floor (b). After the patient is able to raise a hand or knee without pain, it is appropriate to progress to raising the opposite hand and knee simultaneously. Starting the exercise with good form is important to reduce pain and enhance tolerance. Here, a patient is starting in a flexed spine posture (a). Corrections include more lordosis, “life the tail,” and manually

cuing up the rib cage (b). Another poor beginning posture is not having the knees under the hips and the hands under the shoulders (c).



Bird Dog, Beginner

The progression continues with raising one leg or one arm at a time.



Bird Dog, Intermediate

The intermediate bird dog is achieved when the patient is able to raise the opposite arm and leg simultaneously. Raising either the arm or the leg past horizontal should be avoided. The objective is to be able to hold the limbs parallel to the floor for about 6 to 8 seconds. Good form includes a neutral spine with no spine motion and abdominal bracing. Also, here is a cue to push the heel away and make a fist to enhance muscular engagement and reduce the tendency for hip hiking.



Bird Dog, Advanced

To develop the motor control challenge further, the patient should not rest by placing the hand and knee to the floor after each holding repetition. Rather, after extending the normal bird dog position (a), the patient should sweep the floor with the hand and knee so that no weight is borne by either. The patient is shown beginning the sweep (b), at the innermost point of the sweep (c), and coming back out of the sweep (d, e). The patient then extends the active limbs back out into the bird dog position. This technique also allows muscles to reoxygenate with each sweep cycle (McGill, Hughson, and Parks, 2000). Finally, the raised hand and arm should be stiffened with cocontraction including the shoulder. With focus on the upper back, these muscles (thoracic longissimus and iliocostalis) achieve a higher level of contraction. The patient should continue the progression by pushing the heel away. Note how this facilitates high activation in the gluteal and low back muscles on the side of the elevated leg.





Special Conditions

Thoracic kyphosis. As some people age, they develop thoracic kyphosis—a flexed thoracic spine (see [figure 10.9a](#)). However, younger people also develop kyphosis through habitually poor posture, coupled with too much sitting during computer and cell phone use. Some athletes tend to become kyphotic, such as long-distance runners and some back-dominant athletes such as rowers.

The stretch described here was developed by the great Czech neurologist Karol Lewit and based on proprioceptive neuromuscular facilitation (PNF). The clinician is on one knee while the patient adopts a full kneel. The patient's fingers are interlaced behind his head and uses a hip hinge placing his elbows on the clinician's thigh. (see [figure 10.9a](#), b). This is held for about 10 seconds and the 5 pound push is released and the clinician draws the elbow forward extending the thoracic spine (c). The patient can learn to do this on his own, replacing the clinician with a chair (d). Upon releasing the 5 pound elbow force, he draws his buttocks back to his heels.

Figure 10.9 (a) The clinician shows how the fingers are interlaced and placed behind the head and neck. (b) The patient places his elbows on the clinician's thigh while the clinician ensures the patient has a neutral lumbar spine. The patient then pushes down through the elbows with 5 pounds (2.3 kg) of force for about 10 seconds. (c) As the patient releases the effort, the clinician pulls the elbows forward, cuing the motion with the hand over the thoracic region. No force is applied with this hand. This is repeated for perhaps 3 or 4 cycles as long as the thoracic spine releases in each cycle. The clinician should help the patient in standing because he may feel unnerved with his newfound height increase. (d) The patient can be coached to do this on his own by replacing the clinician with a chair or similar object. Upon releasing the tension, the pelvis moves towards the feet to create the thoracic extension.





Arthritic stenosis. As many people age, particularly those who had careers characterized by lots of spine motion (e.g., plumbers, recreational soccer players), they tend to develop arthritic changes in the vertebrae. This narrows the canals for the spinal cord and roots. Longer-duration standing and walking usually causes motor loss. The usual approach is to teach these patients how to stay in spine flexion. We have very good success with those who have failed this approach by doing the opposite. More flexion in daily activity is often counterproductive because the posture results in more spine loading, not less. We provide spine extension drills ([figure 10.10](#)) and corrective interval walking programs ([figures 10.11](#) and [10.12](#)).

Figure 10.10 Extension drills may be simply prone lying (a pillow may be needed under the pelvis).



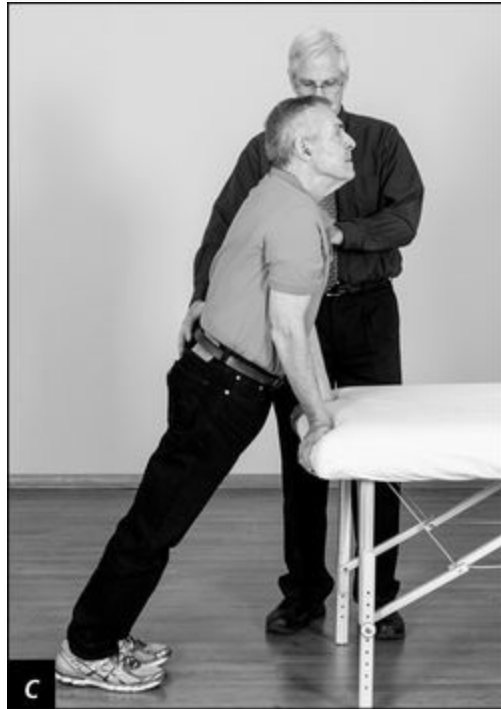
Figure 10.11 This drill prepares the patient with stenosis for walking by assisting him to achieve a more stress-free standing posture. (a , b) Arm walking starts with the patient placing one hand above the other and literally walking the hands up the wall. (c) Cuing the hips to move toward the wall helps the patient achieve the desired upright posture.

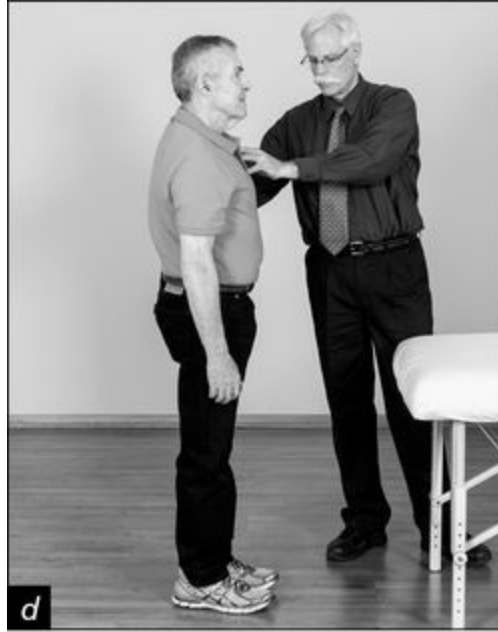




Figure 10.12 (a, b) The patient begins walking, and just as discomfort starts, the patient moves to the strategically placed park bench (or other such object). (c, d) Standing with the hips close to the bench, load is applied down the stiffened arms to add traction to a tolerably extended spine. This posture is held as long as it is comfortable but no longer than about 1 minute. Hopefully, symptoms are lessened or eliminated. Now the patient repeats the walk until the next cycle of pain begins. The patient repeats the training session each day, slowly adding more intervals. This is a technique to achieve pain-free walking in cumulative intervals.







Claudication. Claudication is often confused with stenosis; however, it is a symptom of peripheral arterial disease. In claudication, atherosclerotic plaque buildup narrows the arteries, starving the working leg muscles of oxygen, causing leg pain. Once again, a walking program is recommended (Hirsch et al., 2005) based on tolerable intervals (Brunelle and Mulgrew, 2011). Basically, walking is performed until pain becomes moderate. The patient sits until the symptoms resolve; then resumes walking. The goal is to continue this cycling for up to 60 minutes 3 to 5 times week.

Knee or hip replacement. People who have had knee or hip replacement have difficulty moving to the floor for exercise. [Figure 10.13](#) shows a patient standing, braced on a table or countertop, performing the bird dog.

Figure 10.13 Those with knee and hip replacement can perform the bird dog while standing and braced on a countertop or table. Techniques similar to the floor versions of the exercise are used—for example, fist clenching, heel push, abdominal brace, hand and foot squares. The person must maintain good form with a neutral spine.





Overweight. Unfortunately, reality television shows in which overweight people are belittled and injured influence the public, but provide a poor example. These patients have no doubt been conditioned to fail by many previous weight loss attempts. They need a support structure to comply with the clinical exercise prescription. Success will depend first on their ability to curb their overeating of poor food choices. Then simple exercise progressions are commenced with good form (corrected walking and simple starting levels of the spine-sparing drills). Interestingly, I have not found very heavy people with spine instability because their girth is naturally stabilizing. Thus, we focus more on performing work to enhance endurance and strength in the patterns of movement noted earlier. Core work, one-legged exercises, and stretching are de-emphasized.

A Final Note

Paying attention to detail is the hallmark of the successful clinician–patient relationship. Pain triggers identified using the assessment protocol in chapter 9 must be eliminated. Then the therapeutic exercises appropriately progressed and regressed to match the current tolerance of the patient will establish a decline in symptoms. This will serve many patients well and end their days of pain. Those working with people who desire more athletic abilities will need to read on in the next chapter.